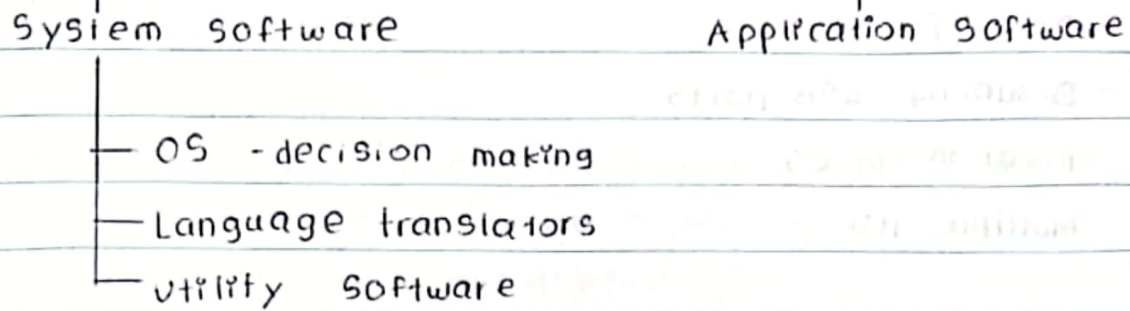
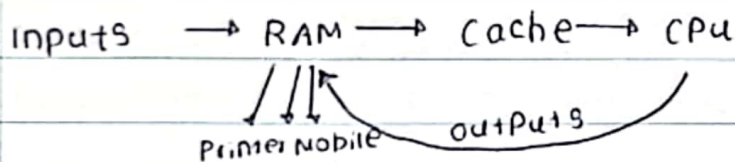


Operating System



Main Aspects

- First software to be installed
- Main Controller and Coordinator



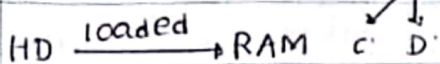
ROM is a firmware. Combination of Hardware and Software
(Not separable)

Note CPU - executor
OS - manager

Bootng Process ROM $\longleftrightarrow$ CPU (ROM, chngement can't be applied)

Accession 9

1. The booting process (file)
2. post (power on self test)
3. Locating the OS (HD, DVD, pendrive)
4. HD $\xrightarrow{\text{loaded}}$ RAM $\begin{matrix} \swarrow \downarrow \\ C: D: \end{matrix}$



CMOS battery - Change ment can be applied

- Date / Time
- Disabling USB ports
- Location of OS
- Multiple HD

• Main difference between client and server based on classification

Client $\xleftarrow{\text{request}}$ Server OS - Sun Solaris
Computer $\xrightarrow{\text{respond}}$ computer - Windows

- linux

- Client computer - Less space

- ubuntu

- Server computer - more space

- unix

Control and Coordinate

- Hardware
- other software
- user

User $\rightarrow$ OS

- Login , not allowing unauthorized access
- allow or disallow to install software
- Compatibility with OS - cpc & OS (32bit/64)

- RAM capacity

- Free disk space

- VGA

- OS is an interface between hardware and user

- Messages for user

Messages - notification
- reminder
- error

- Confirmation - if a user accidentally delete something
- when we leave a software the message to save or discard
- Consistent environment with software (with common interface)

Other software → OS

- Compatible with OS
- To access files (default apps)
Example - open docx in power point
- updates
- Organizing hardware resources (microphone, printers, webcam)

hardware → OS

- Installing hardware
 - compatible with - CPU
 - OS
 - device driver
- OS → some utility software
- Play N play
- Allocating hardware to user and software
- Partitioning Hard Disk
- Formatting Hard Disk, USB, drivers
 - utility software
- Disk scanning
 - virus
 - damaged areas

Logical drivers

OS based on interfaces

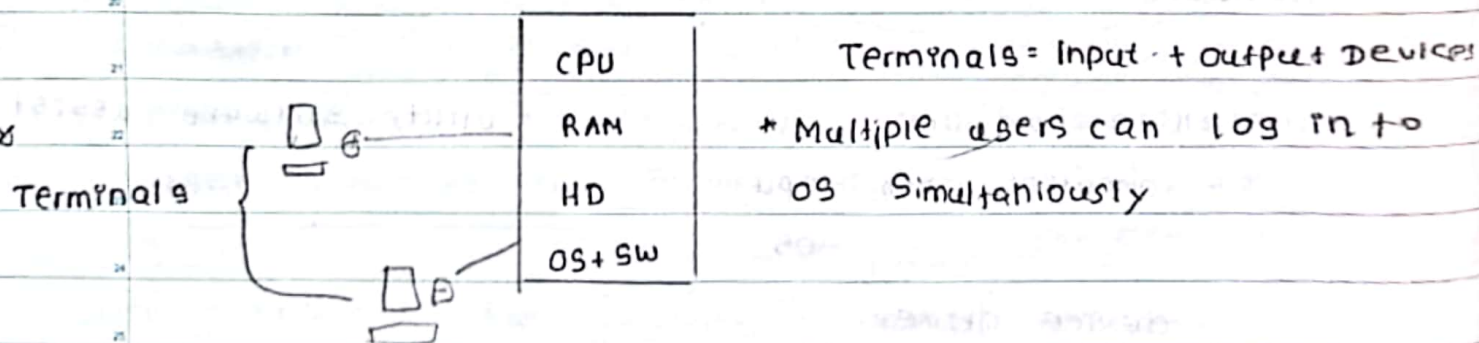
- CLI - Command Line Interface (MSDOS) Command Prompt
- GUI - Graphical User Interface

Note: DOS - Disk OS

OS based on users and task

- 1 Single user and single task (MSDOS), single core CPU
- 2 Single user multitasking - OS, Multiple cores CPU
- 3 Multi user - Multi tasking
Not only the OS, CPU # cores also affect.
- 4 Real time - Processing of the responses are immediate

Mainframe / Mini computers



- In the terminals no software has installed.

~~Processing~~ on

Memory management

Primary Memory refers to primary storage

Primary Storage - RAM

cache

Register

- Memory management is required since it's memories limited on a computer

Content stored in RAM - Inputs

- Data
- Instructions
- Background programs
- outputs
- Launched software (copies of the software)
- opened files

cpu cannot directly access harddisk.

- OS

- Malware

Symptoms of filled up RAM - slow performance

- device get stuck, restart
- battery drain

the usage of RAM
Before these symptoms come up, we can check task manager through task manager.



These processes are organized by the OS.

- When RAM is filled up and the user is unaware, user input data will go store in the ~~that~~ free space of hard disk temporarily. This is called "virtual memory". When we partition the HD we should leave free space.
- CPU we can't directly access HD. Data of HD should come to RAM. When the RAM and HD both filled up, swapping can happen.
- Swapping is managed by OS. Swapping will make computer slower.

• CPU is the executor.

Memory Management

- Memory space tracking
 - virtual Memory
 - memory location - memory address
- Virtual (logical) Addresses
Physical Addresses

Logical Address = Address shown to user by the OS

Virtual (logical) Address = Physical Address = Actual Location of the RAM

On memory there are exists physical locations.

But the user is unaware of the physical locations.

Therefore OS provide virtual addresses for the understanding of a user. Similar examples is found with hard disk.

Examples - Partitions

folders

- virtual Addresses are for users.
- Physical Addresses are using for retrieving, storing

of virtual addresses > # of physical addresses.

32 bits } word size
64 bits }

• Because of the virtual memory.

When we install 64 bits OS Computer should be compatible with
Background process - RAM
- CPU architecture

When we install RAM in 32 bit computer, because standard size is 4 GB

speed of memory (RAM)

speed 1600 MHz — Speed of travelling inputs from RAM to CPU

The speed or force of the overall processing of the computer

CPU — System bus — RAM — CPU 64 -

L1 cache - Register Space

L2 & L3 - cache - Cache

Cores

logically having 24 processors. Together each core can do separate task simultaneously.

Process / P ID - CPU identity software using P-ID

System Interrupts - mouse movement

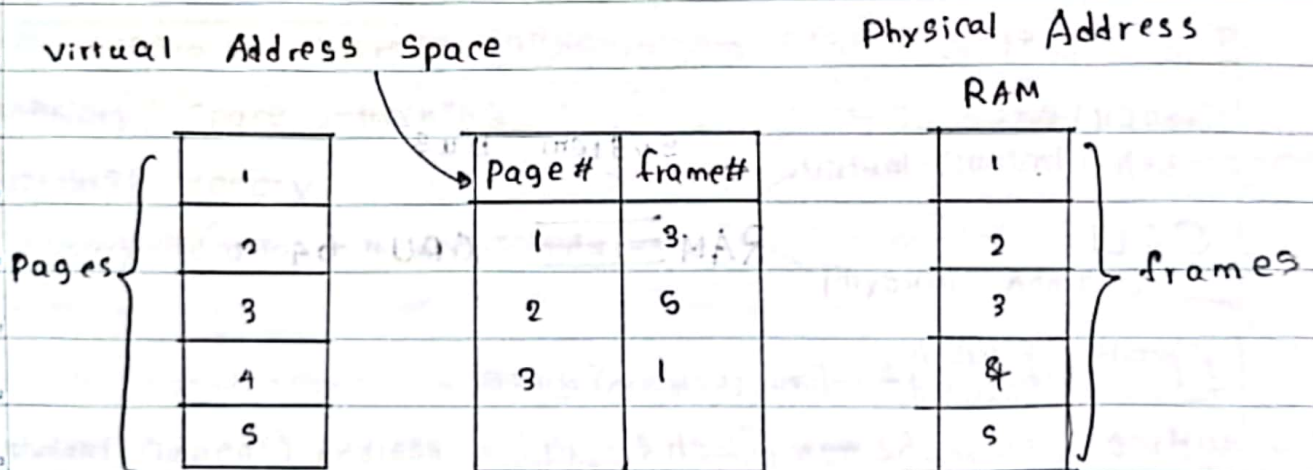
Startup Application automatically open when we switch on the computer.

Startup Impact - time take to start work after

2022.10.05

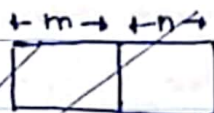
Virtual address $\xrightarrow{\text{Mapped}}$ Physical address

This process done by hardware. And the process is managed by MMU (Memory Management Unit). - close to CPU
firmware



page size = frame size

of pages > # of frames $\rightarrow$ Because of free Space in Hard Disk.



Page # page 2

Pages

* Fixed size

* Page size is 2^n bytes

$$n = 8 \text{ bits}$$

$$2^8 = 256 \text{ bytes}$$

$$2^n = 15$$

$$2^{15} \text{ bytes}$$

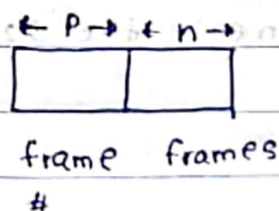
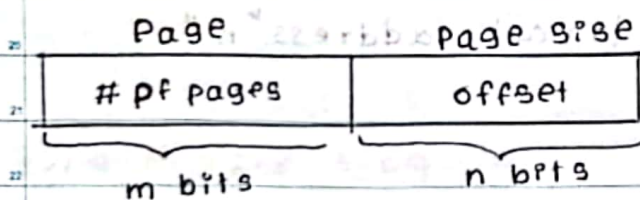
$$2^{10} \times 2^5 \text{ bytes}$$

$$2^{15}$$

$$32 \text{ KB}$$

Page size is determined by number of offset bits used which is represented by "n". (Representation)

Size of page is expressed by bytes.



$$2^m = \# \text{ of pages}$$

$$2^n = \# \text{ of page size}$$

eg.: $m = 7 \text{ bits}$

$$2^7 \longrightarrow 128 \text{ pages}$$

$$2^m > 2^n$$

$$\# \text{ pages} > \# \text{ frames}$$

But the size of both page and frame is equal

1. MMU has 64 pages. Size of a single page is 512 bytes.

Calculate the number of bits required for the logical address space.

$$2^6 = 64$$

$$n = 6$$

$$\# \text{ of pages} = 64$$

$$\text{Size of a page} = 512 \text{ KB}$$

$$2^n = 64$$

$$2^n = 512 \text{ KB}$$

$$2^m = 2^6$$

$$2^n = 2^9$$

$$m = 6$$

$$m + n = 9$$

2. Page size 2KB. MMU can maintain a maximum of 64 pages.

Determine the size of logical address.

$$\# \text{ of pages} = 64$$

$$\text{Page size} = 2 \text{ KB}$$

$$2^n = 2 \text{ KB} \times 2^{10}$$

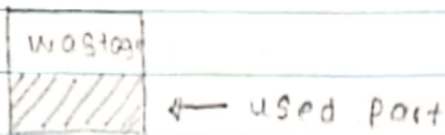
$$2^n = 2^{11} \times 2^{10}$$

$$2^n = 2^{21}$$

$$n = 21$$

In physical address space of 64 is 32 bits or 6

Page size $\uparrow$, # of pages $\downarrow$ - The problem of this is waste



Page size $\downarrow$, # of pages $\uparrow$ - This one is better. But there is problem. ∇ Difficult to handle due to many pages. (more work for OS)

MMU has 2^{16} pages for mapping. A single page is 512 bytes.

Calculate the number of bits required for logical address space

Size of a page = 512

of pages = 256

$$2^m = 256$$

$$2^n = 512$$

$$2^n = 2^9$$

$$n = 9$$

to convert KB $\rightarrow$ bytes

$$m = 8$$

Size of the logical address = 8 + 9

Size of the logical address = 17 bytes

of pages $\times$ size of page = number of bits = 17 bits.

$$216 \times 512$$

$$2^9 \times 2^9$$

$$2^{17} \text{ bytes}$$

require for logical

address space

Size of a single page is 2KB. MMU handles 128 pages.
Calculate the number of bits required for logical address space.

$$2 \times 128$$

$$2^1 \times 2^{10} \times 2^7 \text{ --- converting bytes to bit}$$

$$2^{1+10+7} \text{ bytes} = 2^{18} \text{ bytes} = 256 \text{ KB}$$

$$\text{Total of } 2^{18} \text{ bytes}$$

$$\text{Now } 2^{18} \text{ bytes} \times 8 \text{ bits} = 2^{21} \text{ bits}$$

$$= 2097152 \text{ bits}$$

Size of a page is 1KB. There exist 64 frames.

I Calculate physical spaces of RAM

II Calculate # of bits used in physical Address Space

$$\# \text{ of frames} = 64$$

$$\text{Size of a frame} = 1 \text{ KB}$$

$$2^p = 64$$

$$2^n = 1 \text{ KB}$$

$$2^p = 2^6$$

$$2^n = 2^{10} \text{ bytes}$$

$$p = 6$$

$$n = 10$$

$$=$$

$$2^6 \times 2^{10}$$

$$2^{16}$$

$$=$$

$$\text{frame size} = \text{Page size}$$

$$= 1 \text{ KB}$$

$$= 2^{10} \text{ bytes}$$

$$\# \text{ of bits in Physical}$$

$$\text{Address Space} = 16 \text{ bits}$$

$$= \# \text{ of frames} \times \text{frame size}$$

$$= 64 \times 2^{10} \text{ bytes}$$

$$= 2^6 \times 2^{10} (1 \text{ KB})$$

$$2^{10} = 1 \text{ KB} \text{ due to}$$

$$= 2^6 \times 64 \text{ KB}$$

$$2^6 \times 64 \text{ KB}$$

$$1 \text{ MB} = 1024 \text{ KB} \\ 2^{10} \text{ KB}$$

There exist 256 frames. Size of frame is 4KB.

i Calculate the RAM Space.

ii Write down the # of bits in physical Address Spaces.

of frames $\times$ frame size

256

$$\times 2^2 \times 2^{10}$$

Not clear

2⁸

$$\times 2^{12}$$

$$2^{20} \text{ bytes} \left((2^{10} \times 2^{10}) \text{ bytes} \right) \text{ --- } 1 \text{ KB} \text{ --- } 1 \text{ MB}$$

$$20 \text{ bits} \times 1 \text{ KB} \times 1 \text{ KB} \times 1 \text{ KB} = 1 \text{ GB} \\ 1 \text{ KB} \times 1 \text{ KB} = 1 \text{ MB}$$

Page size is 8KB. There exist 256 frames.

i Calculate RAM Space.

ii Calculate # of bits used in physical Address space.

of frames $\times$ frame size

256

$$\times 2^3 \times 2^{10}$$

2⁸

$$2^{13} \times 1 \text{ KB} \times 1 \text{ KB} = 1 \text{ MB}$$

2²⁴ bytes.

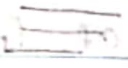
$$2^4 \times 2^{10} \times 2^{10}$$

16 MB

$$16 \text{ MB} \text{ --- Page Size} = \text{Frame Size}$$

of bits in physical = 24 bits

Address Space

RAM  CPU 64/32 (word size) architecture

CPU should handle 64 bits simultaneously. Through system bus transmit data from RAM to CPU and vice versa.

System Bus 64 but CPU architecture 32 bits. In this case unused lanes are there. But it will work.

System Bus & CPU architecture Both should be same bits. Previously before 32 bits, 16 bits used.

A computer is having a 32 bits of OS. Calculate maximum RAM capacity that could be supported.

32-bit OS $\therefore$ RAM Capacity = 2^{32} bytes

32-bit OS $\therefore$ RAM Capacity = 2^{32} bytes

$$2^{32} = 2^8 \times 2^{10} \times 2^{10} \times 2^{10} = 1 \text{ GB}$$

4 GB RAM

User has plugged in 8 GB RAM but when checked on system properties the amount of RAM space is 4 GB. Explain the reason for this scenario.

On the computer OS is 32 bits. 32 bits OS can only plugged in 4 GB RAM. 64 bits OS, the maximum size of RAM is 8 GB.

Even though we plugged 8 GB RAM to a 32 bit OS, it only recognize 4 GB. There for compatible OS is

32-bit OS	64-bit OS
$2^8 \times 2^{10} \times 2^{10} \times 2^{10}$	$2^8 \times 2^{10} \times 2^{10} \times 2^{10}$
4 GB	8 GB

idle hand

No.

Date

64 bits

2⁶⁴

$2^4 \times 2^8 \times 2^{10} \times 2^{10} \times 2^8 \times 2^8$

16 HB

2022.10.30

Evolution of OS

① No OS (1940's, 50's)

- computer had only firmware and hardware.
Hardware + software $\rightarrow$ firmware
- Single user system. Only one person can use at one time.
- Only one program at a time can handle. (serial one time process).
- users, ~~data~~ were directly accessing hardware using Machine language.
- CPU idle ~~time~~ time is high.

② Batch processing (1950's & 1960's)

- To process, the set of work has to be collected.
- Set of work should be similar.
- CPU idle time is high.
- Magnetic tape used as secondary storage.

- for the 1st time OS became an interface between user and hardware. In other words, user cannot directly access hardware.

- uniprogramming (Only one program at a time)

③ Multi-~~processing~~ Programmed batch processing

- Begin OS modern OS.

- Introduced with 3rd generation computers.

- Multiple program can be stored in the RAM. Concept of modern OS started in this era.

between CPU can swap with programs

- CPU idle time is low.

④ Time Sharing system

- Era of modern OS.

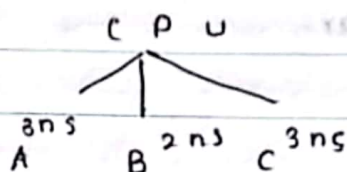
- CPU handles multiple programs at the same time. Each program is given a kind slice of time.

$$s = 1000 ms \quad 1s = 10^9 ns$$

$$ms = 10^3 \mu s$$

$$\mu s = 10^3 ns$$

$$ns$$



CPU switching between tasks.

This is called time slicing.

There are separated times for each tasks.

In mainframe and mini computers use time slicing. Therefore, CPU switches between processes based on the time slices given. This is known as context switching.

- To move between processors, ^(additional) overhead time is used.
- Due to time sharing, user interaction can be ~~input~~ improved.

Process Management

- Program is a set of instructions for the CPU to execute.
- Program that's active/running known as processes.
- Some processes are activated by user and some are activated by OS.

Types of processes

- ① I/O bound - for the ^{completion} of the process input and output.
 - ② Processor bound - doesn't rely on user inputs. Example - Virus Guard.
- Relies on user inputs/output

Attributes of a process

- PID - Process ID
- State - Ready, Run, Blocked
- Priority Level - different levels of priorities.
- Program Counter. The next instruction to be handled
- Input, output information
- Accounting information
- Memory location

How does it uses resources in OS.

- Request a resource
- Use the resource
- Release the resource

4 types of Deadlocks

- Mutual Exclusion

If one of resources is shareable, no deadlock.



- Hold and Wait

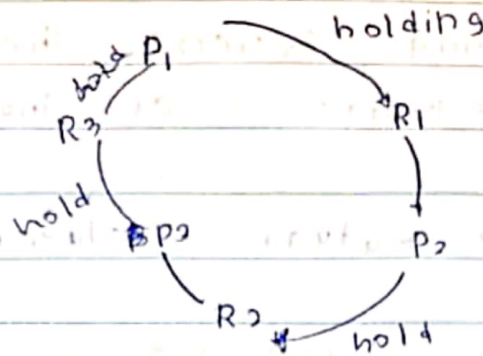
A process is holding at least one resource and waiting for resources.

- No Preemption

A resource cannot be taken from a process unless the process releases the resources.

✓ - Circular wait

A set of processes waiting for each other in circular form. None of the resources are shareable. All resources are stuck.



for
Method, handling deadlock.

- Deadlock prevention or avoidance
- Deadlock detection or recovery
- Deadlock ignorance

Prevention can be done

Resource utilization is low in prevention and avoidance.

With older operating systems there were no deadlocks. But what was the limitation of those operations.

Explain the terms mutual exclusion using a suitable diagram.

Explain difference between preemption and non-preemption.

List down several hardware and software resources that processes will compete for.

efficiency is low
utilization is low
Complexity and time consuming is high.
Limited scalability



3 Preemption

A resource can be taken from a process ^{at} ~~when~~ process releases the resources any time.

None preemption.

A resource can't be taken from a process ~~when~~ unless process releases the resources

4 Hardware

- RAM

Printers

CPU

Software

- Set folder

files

Softwares

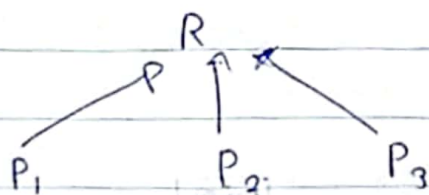
Deadlock vs very little ignorance

If deadlock is happen in windows, and unix, then let it happen and reboot the system

Deadlock detection or recovery - Solutions

- Preemption

- Preemption is time sharing



Each process utilizes a resources as follows.

Process request.

- Request - from OS
- Use - grant the resource or block.
- Release

- In preemption, second & third stages will be changed. Releasing will happen several times.

- Non preemption, releasing happen only one time.

Example of resources - files

- CPU Cycles
- memory space
- i/o devices

Resource - Allocation Graph

Process



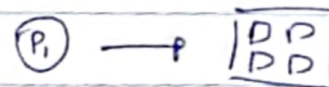
Resource Type with 4 instances



- How many resources can be shared to the process.

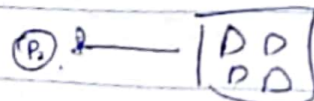
P_1 request

P_1 requests instance of R_1



- Process request 4 instances

P_2

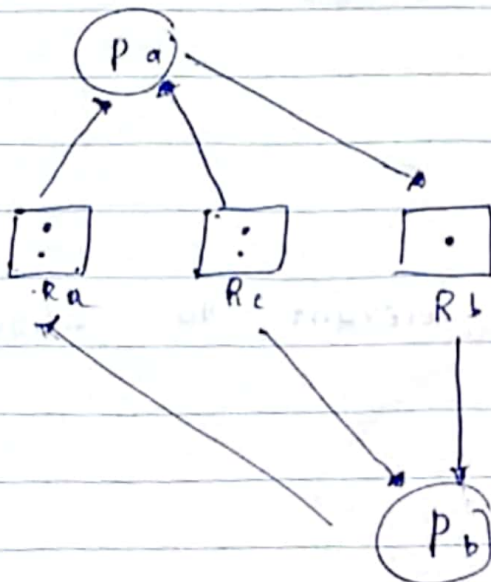
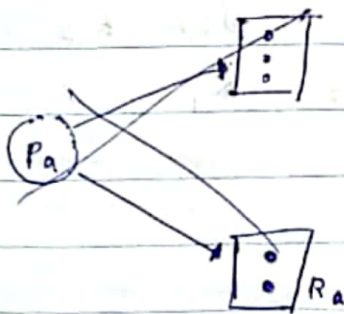


Draw a resource allocation graph for the following scenario.

1. There are 2 processes namely P_a and P_b . There are 3 resources R_a, R_b, R_c . R_a has 2 instances, R_b only has 1 instance and R_c has 3 instances.

— Currently P_a is holding onto an instance of R_a and R_c . Now P_a is requesting for an instance of R_b .

— P_b is currently holding onto an instance of R_b and R_c . Now P_b is requesting for an instance of R_a .



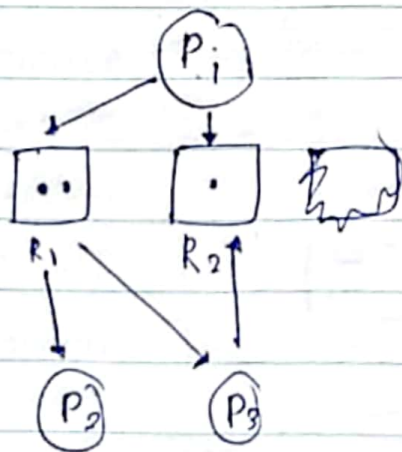
Point out a problem associated with the diagram

- Even though P_a is requesting an instance of R_b , it is already holding on by P_b .

Draw a resource for the following scenario

There are 3 processes, P_1, P_2 and P_3 . There are 2 resources R_1 with 2 instances and R_2 with only one instance.

- Currently P_2 and P_3 are holding onto R_1 .
- P_1 and P_3 are requesting for R_2 .
- P_1 is also requesting for R_1 .



Identify any issues relevant to above scenario.

Instances of R_1 are held by P_2 and P_3 .
But P_1 is requesting for R_1 .

P_1 is requesting instance of R_2 resource.
But P_3 already holds onto R_2 .

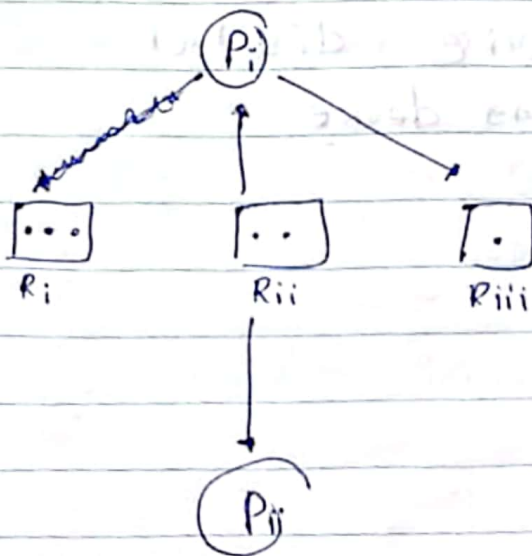
Draw a resource graph for the following scenario

There are 2 processes: P_i and P_{ii} . Currently
there are 3 resources: R_i , R_{ii} and R_{iii} .

R_i has 3 instances. R_{ii} has 2 instances. R_{iii} has 1 instance.

Currently P_i and P_{ii} are holding onto R_{ii} .

P_i is making a request to access R_{iii} .



identify any issues related to above scenario

✓ R_i has no request utilization. Both P_i or P_j not requesting or holding onto R_i .

No issues ~~in~~ here no in this scenario.

2023.11.27

- Digital computer — data is in binary format

- Analog computer - Continuous

- Between 1 and 0, in between no values

- sense only natural property

- weather Prediction

- Sounds, lights

- Hybrid computers — both aspects are there

Example ECG - Heart Beat is analog & Bus in the device digital

- Fuel pumping device

- ECG



11

More in third Generation

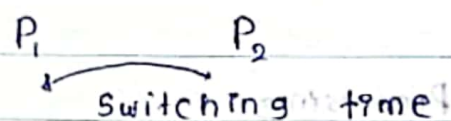
- Time Sharing (CTSS)

Each process has particular period of time

- Multics

- Unix

- Linux



ions

Overhead - not directly involve with process:

- PCs - have client OS

- Server devices - Network OS - ubuntu

Linux

Unix

Sun Solaris

windows Server 2019, 2022

- Distribute - Decentralized Setup

~~Maintain~~ Several

Centralized Server - Maintain Several is Easy

- Applying security measure

- Server should able to

handle

Central Point of Failure

Decentralized advantages - failure of a server,
Distributed Only affect of a part -
only a section of a
organization.

- workload is shared.
- Maintaining is taking more time.

Natural Language Processing

Converting natural language or sound to
Machine Language is called Natural Language
processing.

- Challenges - Frequency of voices.

Process Scheduling

Every program

Process are currently running / executing is called
as process.

- A program may have several instances. Same
Program can have several instances, or thread.

- Text category is source code we type.

- Data class is variables.

- Heap is whatever the changes we do to the
code.

The Space in the memory in

Stack - is for local variables is called Stack

Low address - Starting point of memory allocation for the process.

High - Ending point of memory allocation for the process.

Difference between low and high addresses are memory occupied by the process.

Process states in Operating System

New (create) - The moment when process is created

Ready - ^{Process} waiting for instances

Run - The process handle at the moment by CPU is

R _e	R _e	R _{un}	R _e	R _e
P ₁	P ₂	P ₃	P ₄	P ₅

process might be changed Run to Ready State and Ready to run state multiple times

Block or Wait - When user inputs are not input in the particular time, access will be blocked

Waiting is can happen all the time considering nano seconds.

Example - two computers are giving order to print with

Terminated or completed: - Once a process is finished, process will be terminated.

- When forcefully process is stopped.

process will be terminated.

Suspend Ready - When RAM is filled up, then process that is in ready state which is RAM, moved to hard disk.

Suspend wait or Suspend blocked - Process, which is in block state, move to hard disk called suspend wait or suspend block.

location is ready but,